

Development and Test of an Occupational Taxonomy Based on Job Characteristics Theory

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The use of occupational classification systems has enjoyed a long tradition in vocational behavior. This study attempted to develop a new classification of occupations based on the degree of variety, identity, autonomy, significance, and feedback—major job characteristics often identified with the motivational potential of jobs. Based on two randomly drawn samples from 1515 respondents in the *Quality of Employment Survey, 1977: Cross-Section* (R. Quinn & G. Staines, 1979, Survey Research Center, University of Michigan, Ann Arbor, MI), a cluster analysis revealed 10 relatively homogeneous clusters of jobs across both samples. Additional analysis suggested further support for the taxonomy when between-cluster differences appeared to explain the relationship between the job characteristics and job satisfaction, organizational size, and prestige while within-cluster effects for these same variables were nonsystematic. Interpretation of the taxonomy centers on differences between levels of variety and autonomy versus different levels of identity, significance, and feedback across the 10 profiles. The taxonomy is also interpreted using the 15 most frequently occurring occupations in the study sample. Finally, the counseling implications of the results are discussed; limitations of the study are noted and the need for additional research is suggested.

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The development and use of occupational taxonomies has enjoyed a long tradition in vocational psychology (Dawis & Lofquist, 1984; Gottfredson, 1982; Holland, 1985; Suziedelis & Lorr, 1973). The focus of interest for the present study concerns classifying jobs whose work content varies in motivational potential. From the perspective of vocational behavior, knowledge about the motivational potential of various jobs at a more aggregate level of analysis when combined with information about traditional interest and skill requirements can be useful in providing a more comprehensive picture for placement and counseling purposes.

Research to date on the motivational potential of jobs has resulted in several lines inquiry. In one body of work, Dawis and Lofquist (1984) have developed a theory of work adjustment which builds on the concepts of stimulus, response, and reinforcement. A second stream of research, based on job characteristics theory, focuses on the intrinsic motivation generated within individuals by a job's task content (Hackman & Oldham, 1980). The following is a review of these approaches as they apply to the description of jobs.

Work Adjustment Theory

According to Dawis and Lofquist (1984), work environments can be described in terms of both skill requirements and behavioral reinforcers. How individuals adjust to work environments appears to be based on two major premises: (1) that the individual is a responding organism (Betz, Weiss, Dawis, England, & Lofquist, 1966) and (2) that a basic motive underlying work behavior is to seek and maintain correspondence between needs and reinforcer systems. In this view, stimulus conditions in a work environment which are responsive to individual needs not only signal appropriate behaviors but also help maintain the response and the likelihood of future responses. Further, work environments are described in work personality terms called occupational reinforcer patterns (ORPs). A program of research has been underway to develop an occupational classification system based on some 20 reinforcers which appear to reflect three major dimensions: self-reinforcement, environmental reinforcement, and reinforcement-via-altruism (Dawis, Lofquist, Henly, & Rounds, 1979/1982; Shubsachs, Rounds, Dawis, & Lofquist, 1978).

For the present study, a point of departure from the work adjustment approach is to focus on the *intrinsic* or self-rewarding aspects of a job. Crites (1969) has observed that one of the major criticisms of stimulus-response theories is that individuals strive not so much to satisfy needs or reduce drives but to increase or intensify needs. In other words, one way to understand the adjustment between worker and job is to focus on those aspects which might be self-rewarding for the individual worker. In this way a distinction might be made between job elements which are intrinsically motivating to the worker and those which are extrinsically

motivating (Staw, 1976). This is not to say that extrinsic features of work environments (i.e., comfort) are unimportant or that reinforcement mechanisms have little place in theories of satisfaction and motivation, but rather that the task content required in a job may have important implications for understanding person/job fit. At a minimum, it is suggested here that career decisions by individuals can be influenced by the potential for job stimulation as well as fit with work preferences.

This study also focuses attention on distinguishing between job characteristics and individual needs. Describing work environments and individual needs in a similar way, as in the work adjustment approach, makes it difficult except in the most general terms to identify specific job features relevant to the satisfaction of workers. In job characteristics theory, in contrast, separate job design and individual difference indicators which have led to the use of specific job design interventions (Hackman & Oldham, 1980) have been developed to operationalize person/job fit concepts. In addition, it is possible that the same occupations may vary in their job characteristics because of the way organizations structure work flow arrangements. This study, therefore, also distinguishes between work and occupational environments.

In summary, an attempt is made in this study to focus attention on work environments which might differ in the potential for intrinsic motivation of workers. The framework underlying the approach is based on job characteristics theory which is described in more detail below.

Job Characteristics Theory

Interest in understanding the motivational potential of jobs stems from apparent problems associated with increased specialization of jobs and the accompanying reduction in individual control over the work place which occurred with industrialization. Consequently, early research drew attention to the effects of enlarging the scope or variety of tasks and the autonomy associated with task completion (Herzberg, 1968; Litterer, 1965).

More recently, the conceptual link between job characteristics and individual motivation has been developed in the work of Hackman and Lawler (1971) and Hackman and Oldham (1976). A fundamental assertion in their theory is that workers will be more motivated when they learn that something meaningful or worthwhile has been accomplished by their efforts. Specifically, to be motivating, a worker's job should provide the following:

- (1) The job allows a worker to feel personal responsibility for his or her actions. With greater *autonomy*, a worker gains more control over work efforts which, in turn, enhances responsibility that may result from any successes or failures.

(2) The job involves performing intrinsically meaningful or worthwhile tasks. Three ways this might occur are through increased *task variety* where a worker utilizes many different skills to accomplish a task. A worker's job might also focus on an entire unit or service performed as opposed to just a portion of it. Thus, a job's *task identity* may influence the meaningfulness of a worker's effort. Meaningfulness might also be enhanced through *task significance* when a job has a substantial impact on other people.

(3) The job provides information or *task feedback* about how well a worker is performing. Knowledge of results is an important way by which a worker determines how worthwhile his or her efforts have been.

Finally, consistent with a person/job fit approach, it is also posited that individual growth need strength could influence the relationship of job characteristics to various job outcomes (i.e., performance, satisfaction, and absenteeism).

Research on job characteristics has focused on (1) testing for the predicted motivational outcomes; (2) assessing the psychometric properties of job characteristics measures; and (3) identifying potential determinants of job characteristics. While early tests of the approach seem to suggest its usefulness in predicting the satisfaction and motivation of workers, weak to moderate support has been found for individual difference moderators such as growth need strength (Miner, 1980). Questions about the psychometric properties of job characteristics measures center on the problem of dimensionality and the use of self-reports to describe a job (Aldag, Barr, & Brief, 1981; Roberts & Glick, 1981). The use of self-reports appears to be an important issue because, unlike research on work adjustment theory which has used the occupation as the unit of analysis, job characteristics studies have traditionally employed the individual's self-report of the job as the unit of analysis. Regarding dimensionality, studies seem to suggest that skill variety and autonomy tend to empirically overlap (Aldag et al., 1981; Miner, 1980). The dimensionality problem, however, may be reflective of a more general issue of why job characteristics vary across jobs. In this regard, the limited research on the determinants of job characteristics has emphasized the technology of work flow arrangements which might characterize different organizations or functional areas within an organization (Clegg, 1984; Rousseau, 1977).

Focus of the Present Study

The present study undertook an exploratory investigation to develop a taxonomy of occupational profiles based on the five major job characteristics. While there is reason to expect that known occupations should vary between each other in all or some of the job characteristics, it is

also possible for differences to occur within the same occupation. Managerial jobs, for example, should be more stimulating—high on all characteristics—to their incumbents than, say, operative type jobs. Nevertheless, some managers or professionals may have more confined work roles (i.e., less autonomy or less identity) depending on the way their tasks are organized (i.e., technology of work flow). Given this concern, this study used the individual worker's description of the job as the unit of analysis or building block to construct a taxonomy. In using this approach, however, it should be pointed out that inferences about the homogeneity of aggregated job groupings in the obtained taxonomy may still be problematic if variation within groups is high. In other words, jobs within a group should be more similar than dissimilar in terms of the level of job characteristics (Roberts & Glick, 1981; Zytowski & Hay, 1984).

In addition, the validity of the taxonomy can be enhanced if measures of employee outcomes such as job satisfaction are positively related to the job characteristics at the same level of analysis. Measures of organizational characteristics might also be tested in a similar way: Larger organizations, for example, may contain more differentiated organizational structures which could limit the ability of a worker to perform a variety of tasks, exercise autonomy in carrying out tasks, or produce a whole or complete product or service. Thus, organizational size should be negatively related to job characteristics variables. In addition, greater prestige should be accorded to people whose job belongs to a job grouping which reflects more complexity (higher levels of the job characteristics).

METHOD

Sample of Jobs

Data from the *Quality of Employment Survey, 1977: Cross-Section* (Quinn & Staines, 1979a) were employed in developing and testing for a job characteristics taxonomy. The Quinn and Staines (1979a) study, based on a random sample of 1515 employed adults (20 + hr per week), was designed to be representative of all jobs and industries in the United States (see Quinn and Staines (1979b) for more details on the specific sampling procedures).

In order to enhance the reliability of the obtained results for the present study, two samples of respondents were randomly drawn from the larger sample of 1515 respondents. Missing data reduced the total sample to 1475 respondents with 733 falling into the first subsample—Sample A—and 742 falling into the second subsample—Sample B. The analyses were performed first on Sample A and then examined for consistency on Sample B.

Measures

Job characteristics were measured by items adapted from the Task Characteristics module of the Michigan Organizational Assessment Package (Quinn & Staines, 1979b). These items, which are similar in content to other well-known measures of job characteristics—the Job Diagnostic Survey (Hackman & Oldham, 1980) and the Job Characteristics Inventory (Sims, Szilagyi & Keller, 1976)—assess the degree of variety, identity, autonomy, significance, and feedback reported by each respondent about his or her job. Based on the coding procedure from Quinn and Staines (1979a), the responses to the statements describing the respondent's job ranged from 1 = Strongly Disagree to 5 = Strongly Agree. *Variety* ($M = 3.62$, $SD = .8$, $\alpha = .77$) was measured by six items (examples: "My job requires that I keep learning new things; I get to do a number of different things on the job"). Two items assessed *Identity* ($M = 3.38$, $SD = 1.0$, item intercorrelation = .29). The items are: "On my job, I produce a whole product or perform a complete service"; and "My job involves only a small part in producing the product or service"—reverse scored). *Autonomy* ($M = 3.42$, $SD = .8$, $\alpha = .76$) was measured with six items; examples of these are: "I have the freedom to decide what I want to do on the job"; and "It is basically my own responsibility to decide how my job gets done". Three items form the *Task Significance* scale ($M = 4.12$, $SD = .7$, $\alpha = .53$; examples are: "A lot of people can be affected by how well I do my work"; and "I feel that most things I do on my job are meaningless"—reverse scored). Finally, *Task Feedback* ($M = 4.10$, $SD = .7$) was measured with a single item: "Even if no one tells me, I can figure out how well I'm doing on my job."

The reliability estimates for some of the job characteristics scales appear low which could attenuate the magnitude of empirical relationships involving these measures. The lower reliability for the task identity and significance scales may be due, in part, to sampling fewer items than the autonomy and variety scales. The average interitem correlations, however, for the identity and significance scales are .29 and .27, respectively; the variety and autonomy scales are somewhat higher at .38 and .37, respectively. Thus, there appears to be a moderate degree of consistency in responses within each of the multiitem scales. In addition, these results are not altogether inconsistent with the reported reliability estimates of other measures of job characteristics. In the Aldag et al. (1981) review of research on Hackman and Oldham's (1976) measure of job characteristics, for example, they report an average α coefficient of .68 ranging from .30 to .83 across 10 studies. Nevertheless, some caution seems warranted in interpreting the results obtained from the present study.

Both self-report and nonself-report measures were examined in relationship to the job characteristics. Specifically, *Organization Size* was

measured with the following single item index: "About how many people work for your employer at the location where you work? I mean all types of workers in all areas and departments." The response categories were: 1 = 1-9, 2 = 10-49, 3 = 50-99, 4 = 100-499, 5 = 500-999, 6 = 1000-1999, and 7 = 2000 employees and over ($M = 3.09$, $SD = 1.9$). The *Prestige* of each respondent's job was determined by using Temme's (1975) index which is based on ratings by the general public of the desirability of jobs. The index is based on 1970 census titles and varies from a low score of 0 to a high score of 96 ($M = 41.71$, $SD = 14.3$). Finally, a five-item index of "facet-free" *Job Satisfaction* (1 = low to 5 = high level) was employed to measure individual job reactions. The five items ($M = 3.70$, $SD = 1.0$, $\alpha = .77$) reflect the respondent's general state of job satisfaction, whether the job measures up to expectations, whether the respondent would recommend it to a friend, and whether the respondent would decide to take it again.

Analytical Procedures

The method to develop and test for an occupational taxonomy was based on a two-step analytical approach. In the first step, a cluster analysis was performed on both samples separately using the five job characteristics. Generally speaking, in cluster analysis, an attempt is made to locate distinct subgroups of cases by calculating the degree of similarity of each case to a given cluster of cases. In Ward's (1963) method—a clustering technique shown to yield greater accuracy than other cluster methods (Blashfield, 1976)—for each step in the algorithm pairs of cases are combined so that the variation of profiles within groups is minimized. Changes in the squared Euclidean distance measure are often used as an aid to determine which step in the algorithm best produces a classification of cases into clusters. Unfortunately, one of the problems traditionally associated with this procedure is determining the correct point for interpreting the most parsimonious cluster solution (Borgen & Weiss, 1971; Everitt, 1979). From another perspective, a cluster solution with too many clusters may prove difficult to interpret or use in practical situations even though there may be minimal within-cluster variation; yet a cluster solution with too few distinct groups may invite greater within- than between-cluster differences. Therefore, the key problem for interpretation purposes is to identify a cluster solution with the fewest clusters that minimizes within-cluster variation.

Given this concern, an additional analysis was performed to determine the degree of between-cluster variance relative to within-cluster variance at each point in the algorithm. Ordinary significance tests can be viewed as invalid for testing differences within and between clusters (Everitt, 1979). Nevertheless, two components of the F ratio—within- and between-cells variation—can still be interpreted using *practical* significance criteria.

Recently, Dansereau, Alutto, and Yammarino (1984) have developed practical significance criteria or magnitude tests which can be used to determine the extent to which an effect is primarily within or between cells of aggregation or neither. They suggest using an E ratio—the between-unit η correlation divided by the within-unit η correlation—for practical significance interpretation. (The squared E ratio adjusted for degrees of freedom equals the F ratio.) An E ratio of 1.00 can be interpreted as equal (50%) within- and between-cell variation in scores. While their criteria were not designed specifically for use in cluster analysis, the E ratio will become smaller as more dissimilar cases are combined into fewer clusters. As mentioned previously, if the objective of cluster analysis in general and Ward's (1963) method in particular is to identify the most parsimonious solution (fewest clusters with minimal within-cluster variation) then a cutoff point is needed to infer greater between-cluster variation for a solution with minimal clusters. This study used as a cutoff point for interpretation an E ratio of 1.30 or 63% between-cell variation which Dansereau et al. (1984), based on geometry, suggest as a minimum level to infer a between-cell effect.

For this study, the average of the E ratios for each of the five job characteristic variables was used to interpret the cluster analysis results. These calculations were made at the point in the algorithm where more traditional interpretive approaches could also be used. Specifically, where sizable increases in the Euclidean distance score occurred, the E ratio was employed to assess the magnitude of between-cluster variance for a particular cluster solution.

Within and between analysis procedures were also employed to assess the relationship between the job characteristics and other variables. In this second step in the analysis, the units of analysis are the occupational clusters developed in step 1. One way to provide evidence of the validity of the cluster solution is to test if (1) the relationship between the five job characteristics and the measures of satisfaction, size, and prestige is *systematic* based on the between-cluster deviation scores calculated in step 1 and (2) the relationship between the variables is *nonsystematic* based on within-cluster deviation scores also calculated in step 1. In other words, the inference of between-cluster differences in job characteristics can be strengthened further if the within-cluster correlations between the variables are unrelated to each other. One note of caution in this regard is that inferences based on within-cluster deviation scores may be problematic when there is range restriction. In this study within-cluster deviation scores comprise 37% of total variation. Thus, relative to the between-cluster deviations (63% of total variation) this represents adequate variation to conduct correlation analysis.

Finally, in this within and between analysis (WABA) approach (Dansereau et al., 1984), a Z test of difference between the correlations can

be computed by converting the correlations to Fisher's Z' scores, performing a subtraction of the between from the within correlations with consideration given to the appropriate degrees of freedom. The formula for the Z test is the following (Dansereau et al., 1984, p. 131):

$$Z_{BW} = Z'_B - Z'_W / \sqrt{1/(N - J - 2) + 1/(J - 3)}. \quad (1)$$

In summary, a key issue of concern in utilizing these procedures is the strength of inferring between-cluster differences in job characteristics. Thus, where within-cluster correlations between measures are small in magnitude relative to the between-cluster correlations, then a stronger inference can be placed that "whole" occupational groups—determined through the cluster analysis—appear to lend greater understanding to the nature of individual job perceptions. Alternatively, if the within-cluster correlations are equal to the between-cluster correlations, or both are nonsystematic, then less confidence can be placed in using the obtained categories (clusters) of jobs. The results of these procedures are presented in the following section.

RESULTS

Step 1 Results

The cluster procedure was performed first on Sample A followed by Sample B. Shown in Table 1 are the squared Euclidean distance coefficients produced by using Ward's cluster analysis approach. Also shown are the increases in this distance coefficient, E ratios, and η^2 coefficients for each cluster solution. Because of the large number of potential clusters (over 700), only the results for the 1- to 20-cluster solutions are shown. Further, it is at this point in the algorithm that sizable increases in the distance coefficient begin to occur as more and more dissimilar scores on the variables are being combined into fewer clusters.

By reading down the columns in Sample A, a large increase in the distance coefficient appears to first occur at the 7-cluster solution (69.61 increase). For Sample B, an increase occurs at the 6-cluster solution (77.86). To further interpret these results, it may be helpful to examine the E ratios. As shown in the table, for Sample A the average E ratio for the 7-cluster solution was only 1.14 and for the 6-cluster solution in Sample B, the E ratio was only 1.07. In other words, just over 50% (56 and 53%, respectively) of the variation in scores across the five job characteristics variables is between clusters. It is only for the 10-cluster solution that the E ratios reach the desired level of 1.30 across both samples. Because of this study's interest in identifying a workable number of job groups which meet statistical standards, the 10-cluster result was judged to be the most parsimonious solution.

TABLE 1
Cluster Analysis Results

Cluster solutions	Sample A			Sample B		
	Distance coefficient	Change in coefficient	E ratio	η^2	Distance coefficient	Change in coefficient
1	2458.57	604.29	—	—	2562.76	678.51
2	1854.28	262.55	0.52	.21	1884.25	295.88
3	1591.73	234.55	0.62	.28	1588.37	253.88
4	1357.18	152.85	0.81	.40	1334.49	122.37
5	1204.33	105.65	0.93	.46	1212.12	91.50
6	1098.68	89.42	1.04	.52	1120.62	77.86
7	1009.26	69.61	1.14	.56	1042.76	49.75
8	939.65	48.50	1.20	.59	993.01	48.03
9	891.15	44.17	1.30	.62	944.98	47.81
10	846.98	43.09	1.33	.64	897.17	38.17
11	803.89	40.79	1.37	.66	859.00	30.77
12	763.10	34.36	1.37	.66	828.23	29.56
13	728.74	28.44	1.48	.69	798.67	29.49
14	700.30	28.12	1.48	.69	769.18	27.11
15	672.18	21.10	1.56	.71	742.07	26.56
16	651.08	19.94	1.60	.72	715.51	21.47
17	631.14	19.89	1.60	.72	694.04	20.09
18	611.25	18.77	1.69	.74	673.95	19.88
19	592.48	17.70	1.69	.74	654.07	17.57
20	574.78	—	1.78	.76	636.50	—

Note. Sample A, $N = 733$, Sample B, $N = 742$. The E ratio is based on the ratio of between to within η correlations averaged across the five job characteristics.

Step 2 Results

In order to perform correlational analysis, within- and between-cluster deviation scores were calculated based on the 10 clusters as units of analysis. It should be pointed out that in this analysis there are different degrees of freedom when testing within- and between-unit effects. For Sample A, there are 733 cases aggregated into 10 clusters; therefore, between-cluster correlations are based on $J - 2$ or 8 degrees of freedom while within-cluster correlations are based on $N - J - 1$ or 721 degrees of freedom. For Sample B, there are also 8 degrees of freedom between clusters and 731 ($742 - 10 - 1$) degrees of freedom within clusters.

Shown in Table 2 are the within- and between-cluster correlations between the five job dimensions and the measures of satisfaction, size, and prestige. An inspection of the within-cluster correlations suggests that overall their magnitude is small. In contrast, the magnitude of the between-cluster correlations is much larger. Nevertheless, given the different degrees of freedom available at this level of aggregation, a Z test of difference between the correlations was performed to determine if the differences are statistically significant. These results, shown in the lower portion of Table 2, indicate that for many of the variables, there is a systematic difference between the between-cluster and within-cluster correlations.

As shown in Table 2 the Z -test results suggest that for individuals with jobs in the clusters with more variety, autonomy, and significance, there is a systematic relationship with job satisfaction based on between-cluster differences. This effect was consistent across both Samples A and B. With regard to organization size, there appears to be some suggestion that only clusters with greater identity in tasks tended to be located in smaller size organizations. Although, for Sample B, the relationship between size and task variety, autonomy, and significance tended to reflect between-cluster differences. Finally, for the prestige variable, the results across both of the samples suggest a between-cluster effect to be more likely as jobs in clusters with greater variety, autonomy, and significance tended to be more prestigious.

Summary of Step 1 and Step 2 Results

Overall, these results suggest that individual descriptions of job characteristics appear to comprise 10 relatively homogeneous groupings of jobs. Attempts to further validate this empirically derived taxonomy suggest that relationships between job characteristics and affective responses as well as organizational and prestige factors failed to emerge as within-cluster effects. Rather, the between-cluster effects—"whole" clusters—appear to better explain the relationships.

To ascertain the degree of consistency between the subsamples for

TABLE 2
Validation Results of Cluster Analysis

Job characteristics	Overall job satisfaction		Organization size		Occupational prestige	
	Sample A	Sample B	Sample A	Sample B	Sample A	Sample B
Within-cluster correlations						
Variety	.13	.10	.07	-.01	.31	.28
Identity	.06	.08	-.10	-.12	.09	.00
Autonomy	.07	.11	-.16	-.15	.03	.09
Significance	.22	.18	-.02	-.04	.15	.07
Task feedback	-.07	.09	-.07	-.12	-.13	-.02
Between-cluster correlations						
Variety	.94	.88	-.41	-.75	.93	.97
Identity	.49	.44	-.86	-.77	.20	.35
Autonomy	.83	.80	-.49	-.81	.83	.76
Significance	.86	.84	-.34	-.62	.80	.88
Task feedback	.25	.37	-.43	-.07	-.07	-.08
Z test of differences						
Variety	4.23**	3.36**	0.96	2.53**	3.52**	4.74**
Identity	1.25	1.03	3.14**	2.37**	0.30	0.96
Autonomy	2.94**	2.60**	0.99	2.58**	3.05**	2.38**
Significance	2.83**	2.74**	0.88	1.80*	2.50**	3.44**
Task feedback	0.50	0.78	1.03	-0.13	-0.16	0.16

Note. For within-cluster correlations, $r \geq .07$, $p \leq .05$; $r \geq .10$; $p \leq .01$. For between-cluster correlations, $r \geq .63$, $p \leq .05$; $r \geq .77$, $p \leq .01$. To infer a between-cluster effect, $Z \geq 1.66$, $p \leq .05$; $Z \geq 2.33$, $p \leq .01$. To infer a within-cluster effect, $Z \geq -1.66$, $p \leq .05$; $Z \geq -2.33$, $p \leq .01$ (see Dansereau, Alutto, and Yammarino (1984) for more details).

* $p \leq .05$; ** $p \leq .01$.

the individual cluster profiles the distribution of responses for each job characteristic in both samples was divided into three equal parts based on the 33rd and 67th percentiles: low, medium, and high levels. Using this procedure, only 5 of the 50 pairs of means from the samples were not in the same level. Given the exploratory nature of this research, this degree of consistency (45 of 50 pairs) between samples was judged to be adequate. Thus, both subsamples were combined into a single sample for interpretation purposes.

Interpretation of the Cluster Results

In order to interpret the 10 clusters in a meaningful way, it may be useful to discuss the cluster results in a larger or more general framework followed by a more specific discussion of each cluster grouping by itself and in relation to specific occupations.

Shown in Table 3 are the means, standard deviations, and a summary rating of the level of each job characteristic within a cluster group for the entire sample. The 10 job-characteristic profiles are arranged in hierarchical level starting with the highest level profile (Profile 1) to the lowest (Profile 10). Generally speaking, Profiles 1 to 4 are in the top tier because they can be considered highest in motivational potential; Profiles 5 to 8 are in the middle tier; and Profiles 9 and 10 are in the lowest tier. An examination of the distribution of jobs across the 10 profiles shows that no one profile dominates the distribution.

A closer inspection of the results suggests two general patterns: (1) Variety and autonomy tend to be the same level and vary from the level of task identity and significance within each profile; (2) profiles with higher and lower levels of the job characteristics tend to have higher levels of feedback. Regarding the first pattern, it is possible, as suggested in previous research (Aldag et al., 1981), that jobs which involve more autonomy or freedom of action will require a greater variety of tasks. Viewed from the perspective of a work flow cycle, autonomy and variety may represent a worker's inputs, that is, how a worker influences a product or service. Identity and significance, in contrast, may be closely tied to the outcomes of effort in that both refer to the product or service itself. In addition, jobs with greater motivational potential (Profiles 1 to 3) may provide more feedback because of their generally greater variety and autonomy. In contrast, at the lower end of the continuum, feedback may be at a high level because there is little freedom (close supervision) or variety in the work, which means that when mistakes are made the feedback may be largely negative in nature.

This general, albeit speculative, interpretation of the results as a whole may also be useful in examining each profile in more specific terms. Starting with the higher level profiles, it appears that each profile has a distinguishing feature. For example, Profile 1 is high in all job charac-

TABLE 3
Means and Standard Deviations for Combined Samples

Profile no.	N	%	Job characteristics									
			Variety		Identity		Autonomy		Significance		Feedback	
			M	Level	M	Level	M	Level	M	Level	M	Level
1	125	8.5	4.39 (0.4)	High	4.49 (0.5)	High	4.58 (0.4)	High	4.68 (0.4)	High	4.78 (0.4)	High
2	205	13.9	4.22 (0.5)	High	2.71 (0.6)	Low	4.04 (0.4)	High	4.49 (0.4)	High	4.25 (0.4)	High
3	150	10.1	3.97 (0.6)	Med/high	4.43 (0.5)	High	3.14 (0.6)	Med	4.60 (0.4)	High	4.60 (0.5)	High
4	245	16.6	4.01 (0.5)	Med/high	4.05 (0.4)	High	4.04 (0.4)	High	4.19 (0.5)	Med	4.04 (0.2)	Med
5	129	8.7	3.47 (0.5)	Med	4.15 (0.3)	High	3.05 (0.6)	Low/med	3.98 (0.5)	Low/med	4.04 (0.2)	Med
6	94	6.4	3.64 (0.7)	Med	3.12 (1.0)	Med	3.12 (0.9)	Med	4.02 (0.8)	Low/med	1.83 (0.4)	Low
7	113	7.7	3.61 (0.6)	Med	1.87 (0.4)	Low	3.06 (0.5)	Low/med	4.06 (0.6)	Med	4.20 (0.4)	High
8	193	13.1	3.26 (0.6)	Low/med	2.90 (0.3)	Low/med	2.99 (0.5)	Low/med	4.08 (0.5)	Med	4.22 (0.4)	High
9	112	7.6	2.48 (0.6)	Low	3.58 (0.6)	Med	2.83 (0.8)	Low	3.24 (0.7)	Low	4.24 (0.4)	High
10	109	7.4	2.28 (0.6)	Low	1.97 (0.5)	Low	2.34 (0.6)	Low	3.19 (0.8)	Low	4.17 (0.4)	High
Total	1475		3.62 (0.8)		3.38 (1.0)		3.42 (0.8)		4.12 (0.7)		4.10 (0.7)	

Note. Standard deviations in parentheses. Med = medium level.

teristics, whereas Profile 2 is low in identity while high in the other characteristics; Profiles 3 and 4 are medium in autonomy and significance, respectively. Apparently, some jobs (Profile 2) involve considerable depth and scope of involvement but the outcomes of a worker's efforts are directed at products or services which are only a small part of the final product. In other words, there are some types of jobs in which no one person can start or complete a task from beginning to end, even though the nature of the work is important. Profile 3 jobs tend to be somewhat the opposite: variety and autonomy are somewhat lower than the relative amount of task identity and significance of the work performed. One possible explanation is that in some work environments organization rules may provide limitations as to what an employee can do on a job even though the entire product or service is still being completed. Finally, Profile 4 jobs may be motivating because of the greater variety of skills used and autonomy in completing an entire product or service. In these jobs, the significance of the tasks is less even though task identity is high. Perhaps because of the lower significance, less feedback is necessary.

Profiles 5 to 8 appear to fall in the middle levels of motivational potential. As with Profiles 1 to 4, their distinguishing features appear to be similar: task identity and significance tend to vary but variety and autonomy are similar within profiles. Specifically, Profiles 5 and 7 differ primarily in task identity (high versus low); the other dimensions tend to be in the medium or low/medium range. Profiles 6 and 8 tend to differ primarily in task feedback; otherwise the other four dimensions are also in the medium or low/medium range. Finally, Profiles 9 and 10 are quite similar in that both entail the lowest levels of motivational potential. While there is high feedback in both—perhaps of a negative nature—both groups have the lowest levels of each characteristic except for Profile 9 which has a medium level of identity. Once again, many jobs seem to differ in this dimension.

Another way to interpret the cluster results is to discuss them in terms of specific occupations. Recall that the results shown in Table 2 suggest a relationship between the job characteristics and prestige. In addition, the 10 profiles appear to comprise a continuum, at least based on the dimensions of variety, autonomy, and significance. Nevertheless, as suggested in the preceding discussion, work environment factors may influence, for example, whether a job has more or less autonomy. Thus, similar occupations may differ in terms of their membership among the various profiles.

In order to examine the distribution of profiles within occupations, a summary was prepared by taking from the present sample the 15 most frequently occurring occupations using the 1970 Bureau of the Census classification. For example, engineers, nurses, and teachers would be included among the professional occupations. In this way an array of

occupations ranging in prestige can be compared to the 10 job-characteristic profiles. The results shown in Table 4 indicate that a total of 677 jobs (45.9% of the total sample) comprise these 15 occupations. The table is further divided into the three tiers of job characteristics—high, middle, and low; percentages for these tiers are also shown.

Although the frequencies are small in some of the cells, the table shows a general pattern where the professional and managerial occupations tend to comprise the top tier of job characteristics profiles whereas craft, clerical, and sales tend to be in the middle tier and operatives in the low tier. (There were too few specific laborer occupations to include in Table 4.) Service jobs, however, appear to be spread out across the profiles. Nevertheless, there are differences within the occupations as to their placement into each of the profiles. The following is a brief discussion of the specific occupations shown in Table 4.

Starting with the professional level occupations, it appears that engineers tend to occupy jobs where the work has low or medium identity and/or significance but there is usually high autonomy (Profiles 2, 4, and 6). Nurses also seem to follow a similar pattern, except that some of their jobs have less autonomy (Profile 3). Teachers also appear to be in jobs with low identity or medium autonomy (Profiles 2 and 3). In the case of teachers, for example, specialization by subject matter may influence the extent to which a “whole” service is performed. Nevertheless, there is a medium to high level of autonomy in this type of work.

A key characteristic of many managerial jobs is their high autonomy (Profiles 1, 2, and 4), even though a number of these jobs seem to entail work which has low identity and/or significance. One explanation is that many managers may coordinate or work with others to get an entire job completed which requires latitude. Sales work also appears potentially motivating, as 44% of salespeople are in the top tier (Profiles 1 to 4). It appears that some sales jobs may require many skills as well as considerable autonomy. Yet, other sales jobs appear to have less motivational potential (32% in the middle tier and 24% in the low tier). One conclusion is that sales jobs, unlike professional and managerial occupations, tend to vary considerably in their job characteristics. This same pattern also holds for many clerical jobs which tend to be spread across the various profiles, although the largest group is in the middle tier (48%).

Interestingly, the mechanic/repairperson and carpenter jobs bear many similarities to the professional jobs in that they require using a variety of skills and seem to have high levels of autonomy. About one-half of these jobs fall into this category. In contrast, operatives jobs, as might be expected, are generally lowest in motivational potential (Profiles 8, 9, and 10). While these jobs may be lowest in motivational potential, they do contain high feedback. Finally, service jobs tend to be spread out across the profiles, except for cleaning and food service work which are among the low tier jobs.

TABLE 4
Summary of Job Characteristics Profiles by Occupations

Occupations	Job characteristics profiles														Total (col. %)
	High tier					Middle tier					Low tier				
	1	2	3	4	Total	5	6	7	8	Total	9	10	Total		
Professionals Engineers	<i>n</i>	2	6	1	4	13	1	2	1	1	5	—	—	—	18
	%	11	33	6	22	72	6	11	6	6	28	—	—	—	(3)
	<i>n</i>	—	6	4	3	13	—	4	—	3	7	—	—	—	20
Nurses	%	—	30	20	15	65	—	20	—	15	35	—	—	—	(3)
	<i>n</i>	2	15	18	7	42	3	6	6	6	21	—	—	—	63
	%	3	24	29	11	67	6	9	9	9	33	—	—	—	(9)
Managerial/administrative Managers	<i>n</i>	31	34	7	47	119	6	7	1	9	23	3	—	3	145
	%	21	24	5	32	82	4	5	1	6	16	2	—	2	(22)
	<i>n</i>	4	5	2	4	15	3	3	2	3	11	4	4	8	34
Salespersons	%	12	15	5	12	44	9	9	5	9	32	12	12	24	(5)
	<i>n</i>	3	6	7	6	22	9	6	6	12	33	5	9	14	69
	%	4	9	10	9	32	13	9	9	17	48	7	13	20	(10)

[illegible]

DISCUSSION

An attempt has been made in this paper to provide preliminary evidence for an occupational taxonomy based on job characteristics. Given the exploratory nature of this study, further research is certainly needed. There are, however, several potential implications of this study for future research and counseling.

A key finding is that individual perceptions of a job's motivational potential in task content may share a community with other jobs. There is some suggestion that the job categories developed in this study appear to be related to organization size, prestige, and job satisfaction. Most important, these effects were obtained based on between-cluster differences; within-cluster effects were found to be nonsystematic. It would seem that more research is needed to understand further the network of organizationally related contextual effects which may underlie the 10 job-characteristic profiles.

From a conceptual perspective, the obtained results suggest a possible distinction in jobs along four major dimensions: (1) variety and autonomy, (2) identity, (3) significance, and (4) task feedback. Regarding variety and autonomy, although earlier research on job characteristics distinguished between depth (autonomy) and scope (variety) of responsibilities (Litterer, 1965), it appears that many jobs combine the two. This finding is also consistent with factor analytic studies which have reported dimensional overlap in these two job characteristics (Aldag et al., 1981). The results of this study tentatively suggest that different factors in work environments may be operating to affect the degree of variety and autonomy versus identity, significance, and feedback. For example, while many managerial and professional level jobs entail high levels of task variety and autonomy, they vary in the level of task identity or significance.

While the results of this study should be viewed as preliminary, one practical implication is that conventional occupational groups can vary within and between each other in the extent and nature of job characteristics. It was found that sales, clerical, and some service jobs vary widely across the profiles whereas professional, managerial, operative, and some service occupations tend to be confined to several of the job groups. The results, therefore, suggest the importance of distinguishing between work environments and occupational differences in job characteristics. In this regard, it will be important in future work to create a nomenclature of job titles to label job groups of similar motivational content. In this way, career counseling might be more easily facilitated through the use of a common language to communicate the motivational potential of various jobs.

The job characteristics taxonomy could be used in conjunction with other work environment assessment approaches when interest lies in

maximizing person/work environment fit. For example, assuming a job is stimulating for an individual (i.e., Profile 1 in this study), pay or other work context factors may not necessarily fit with individual needs. For this reason, the job characteristics approach might complement (rather than replace) other occupational taxonomies such as found in the work of Holland (1985) or Dawis and Lofquist (1984).

An important limitation of this study concerns the cross-sectional nature of the study. Collecting self-report job information from a single source at one time point leaves open the possibility that the reported effects were a result of using a common method rather than reflective of job properties. It should be pointed out, however, that the within-cluster correlations (see Table 2) were generally small in magnitude relative to the between-cluster correlations. In other words, if the results were a reflection of individual self-reports due to a common method and not the occupational profiles, then both the within- and between-cluster correlations would be more similar in magnitude.

In summary, this study provides preliminary support for the proposition that relatively homogeneous job groups underlie individual descriptions of job characteristics. The ability to apply this taxonomy of 10 different job categories to practical situations, however, depends on extensive programs of research. The approach employed in this study may be helpful in further developing this line of research.

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